Introduction to Limits

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.03

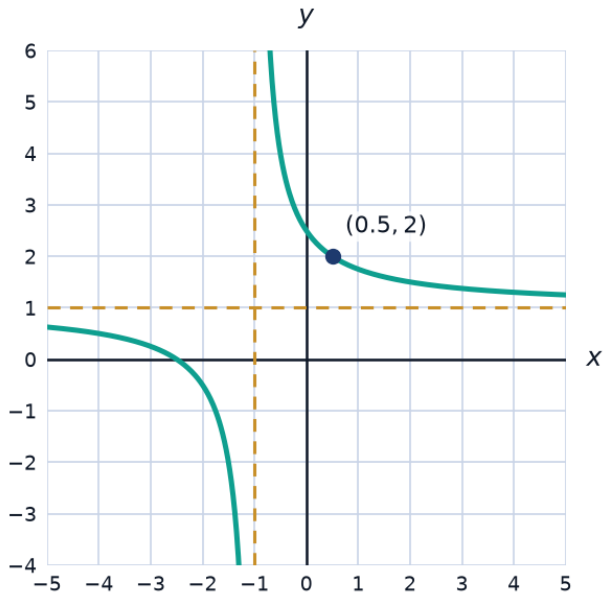
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating limits**

Evaluate $\lim_{x\to 2} \frac{x^2-4}{x-2}$ by first simplifying the expression.

2 **[Foundational]** • **Target: Evaluating limits**

Evaluate by direct substitution where possible:

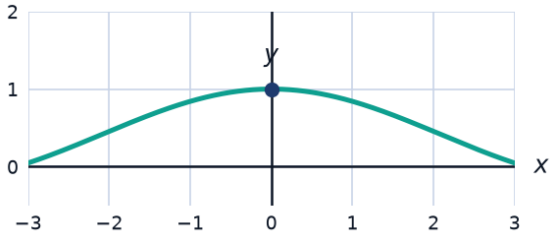


a $\lim_{x\to 3} (x^2 + 2x - 1)$

b $\lim_{x\to 1} \frac{x+5}{x+1}$

3 **[Foundational]** • **Target: Evaluating limits**

Using a table of values near $x = 0$ (in radians), estimate $\lim_{x\to 0} \frac{\sin x}{x}$.



4 **[Foundational]** • **Target: One-sided limits and limits at infinity**

Let $f(x) = \frac{|x|}{x}$. Find:

a $\lim_{x\to 0^+} f(x)$

b $\lim_{x\to 0^-} f(x)$

c $\lim_{x\to 0} f(x)$

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: One-sided limits and limits at infinity

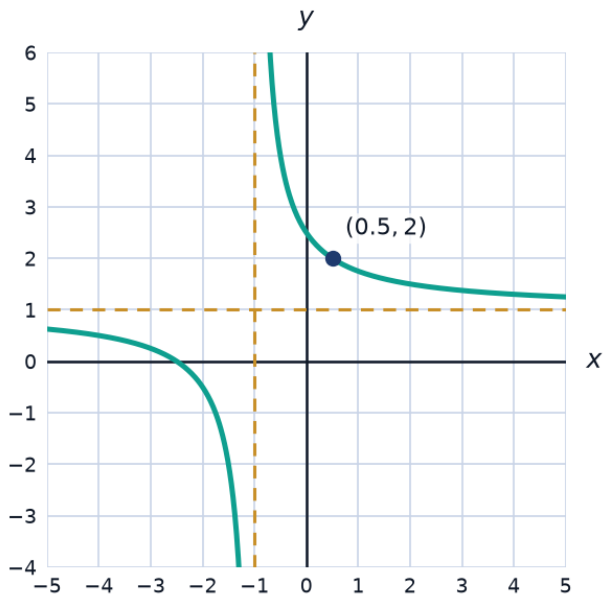
Evaluate each limit at infinity:

a $\lim_{x\to\infty} \frac{3x^2+5}{x^2-2}$

b $\lim_{x\to\infty} \frac{2x+1}{x^2+4}$

6 [Foundational] • Target: One-sided limits and limits at infinity

The graph of $f(x) = \frac{1}{x-2} + 3$ has asymptotes $x = 2$ and $y = 3$. Express each asymptote as a limit statement.



7 [Foundational] • Target: More limit computations

Evaluate $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$ by rationalizing the numerator.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

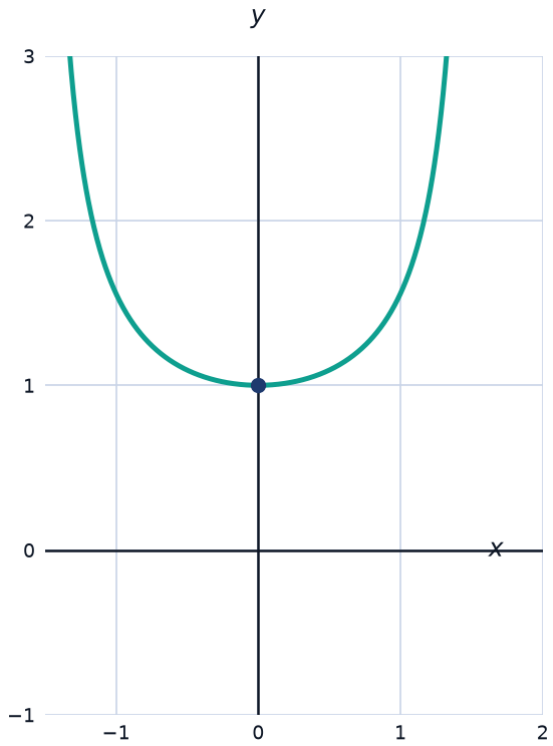
8 [Intermediate] • Target: More limit computations

Evaluate $\lim_{x\to -1} \frac{x^3+1}{x+1}$ by factoring (using the sum-of-cubes formula).

© ClipSAT — clipsat.org | All Rights Reserved

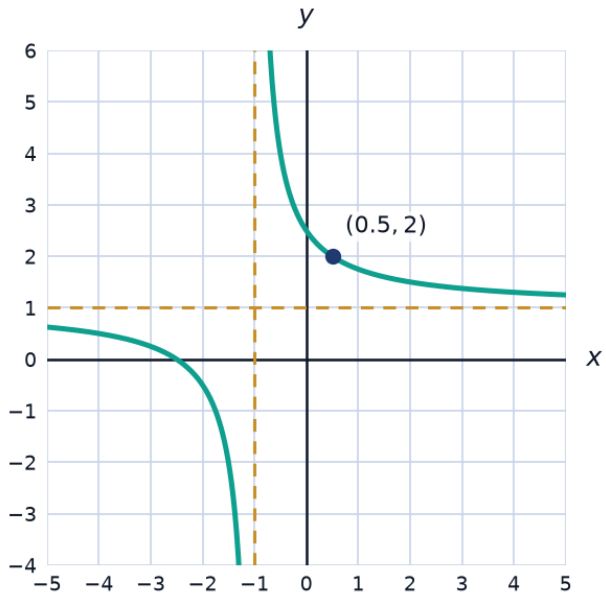
9 [Intermediate] • Target: More limit computations

Evaluate $\lim_{x\to 0}\frac{\tan x}{x}$, using $\lim_{x\to 0}\frac{\sin x}{x}=1$ and $\tan x=\frac{\sin x}{\cos x}$.



10 [Intermediate] • Target: Continuity and limits together

A function f is defined by $f(x)=x+3$ for $x\neq 2$, and $f(2)=10$. Find $\lim_{x\to 2} f(x)$ and state whether f is continuous at $x=2$.



11 [Intermediate] • Target: Continuity and limits together

Estimate $\lim_{x\to\infty}\left(\sqrt{x^2+1}-x\right)$ by rationalizing.

© ClipSAT — clipsat.org | All Rights Reserved

12 **[Intermediate]** • **Target: Limits from graphs and tables**

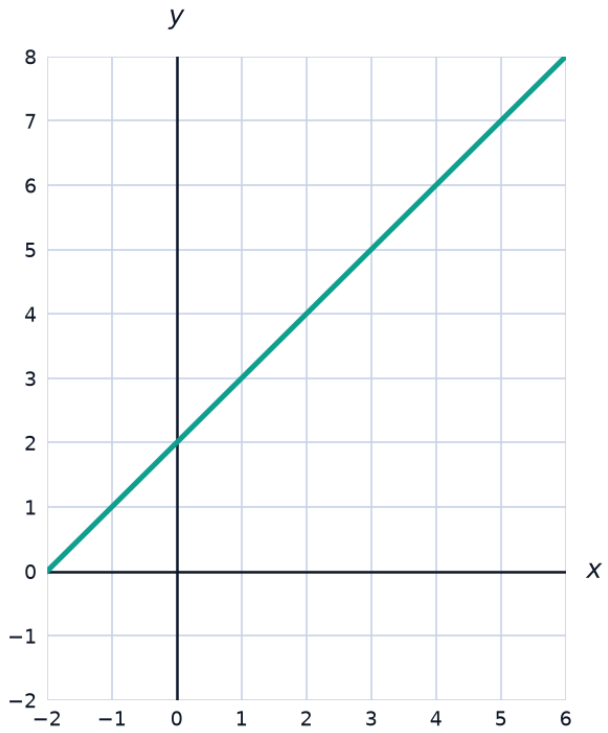
A function g satisfies $g(1.9) = 5.8$, $g(1.99) = 5.98$, $g(1.999) = 5.998$, $g(2.1) = 6.2$, $g(2.01) = 6.02$, $g(2.001) = 6.002$. Based on this table, estimate $\lim_{x \to 2} g(x)$.

13 **[Intermediate]** • **Target: Limits from graphs and tables**

Explain, using one-sided limits, why $\lim_{x \to 0} \frac{1}{x^2}$ exists as an infinite limit (both one-sided limits agree), while $\lim_{x \to 0} \frac{1}{x}$ does not exist even as an infinite limit.

14 **[Intermediate]** • **Target: Visualizing a limit graphically**

The graph of $f(x) = \frac{x^2 - 4}{x - 2}$ (which simplifies to $x + 2$ with a hole at $x = 2$) is shown. Use the graph to estimate $\lim_{x \to 2} f(x)$, and explain why the hole doesn't affect the limit's value.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

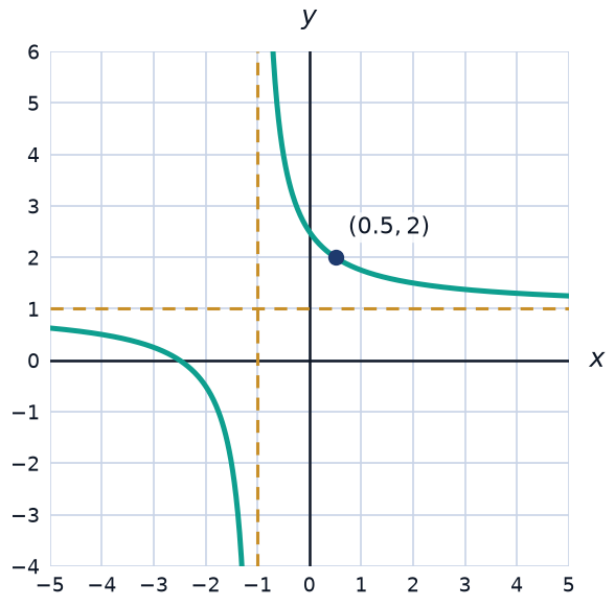
15 **[Challenging]** • **Target: The formal idea of a limit (intuitive)** • **Cross-Lesson: 4.01 Circles and Parabolas**

Explain, in your own words, what it means for $\lim_{x \to a} f(x) = L$, without using formal epsilon-delta language.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Limit laws • Cross-Lesson: 4.02 Ellipses

Given $\lim_{x\to 3} f(x) = 5$ and $\lim_{x\to 3} g(x) = -2$, use the limit laws to evaluate:



a $\lim_{x\to 3}[f(x) + g(x)]$ **b** $\lim_{x\to 3}[f(x) \cdot g(x)]$ **c** $\lim_{x\to 3} \frac{f(x)}{g(x)}$

17 [Challenging] • Target: Limits involving piecewise functions • Cross-Lesson: 4.03 Hyperbolas

A function is defined by $f(x) = x^2$ for $x < 1$ and $f(x) = 3x - 1$ for $x \geq 1$. Find $\lim_{x\to 1^-} f(x)$, $\lim_{x\to 1^+} f(x)$, and state whether $\lim_{x\to 1} f(x)$ exists.

18 [Challenging] • Target: Limits from graphs and tables • Cross-Lesson: 5.01 Vectors: Components and Operations

Consider the rational function $f(x) = \frac{ax^2 + bx + c}{dx^2 + ex + f}$, where a, b, c, d, e, f are non-zero constants. Given that $\lim_{x\to\infty} f(x) = \frac{1}{2}$ and $\lim_{x\to -1} f(x) = \infty$, determine the value of $\frac{a}{d}$.

19 [Challenging] • Target: Visualizing a limit graphically • Cross-Lesson: 5.02 The Dot Product

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(x)$ as x approaches 3? Show your complete algebraic work.

20 [Challenging] • Target: The formal idea of a limit (intuitive) • Cross-Lesson: 5.03 Parametric Equations

Consider the rational function $f(x) = \frac{x^2 - 4}{x - 2}$. What is the value of the function as x approaches 2? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Intuitive Concept and Numerical Estimates

What does $\lim_{x \to c} f(x) = L$ mean intuitively?

a. As x gets arbitrarily close to c (from either side, $x \neq c$), $f(x)$ gets arbitrarily close to L

b. $f(c) = L$ always

c. The slope of f at c is L

d. $f(x)$ is continuous at c

22 [Foundational] • Target: Intuitive Concept and Numerical Estimates

Use numerical estimation with $x = 1.9, 1.99, 2.01, 2.1$ to estimate $\lim_{x \to 2} (3x - 1)$.

a. 5

b. 6

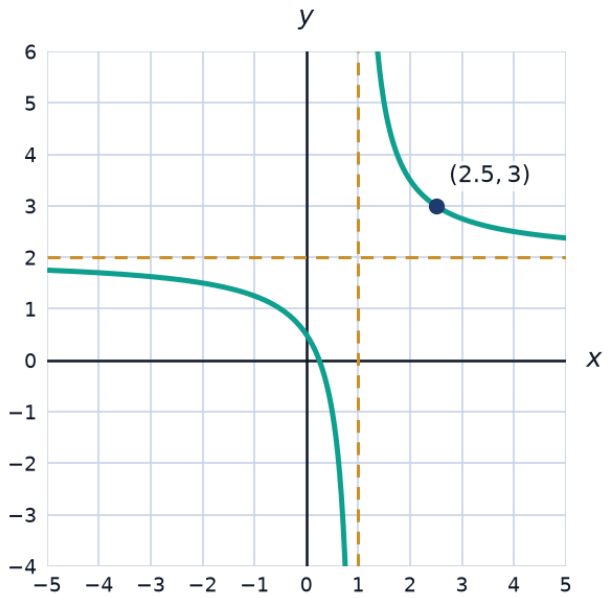
c. 4

d. 3

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Intuitive Concept and Numerical Estimates

If $f(3)$ is undefined, can $\lim_{x \to 3} f(x)$ still exist?



a. Yes, because limits describe behavior near $x = 3$, not at $x = 3$

b. No, limits require $f(c)$ to be defined

c. Only if $f(x) = 0$

d. Only for polynomials

24 [Foundational] • Target: Graphical Limits and One-Sided Limits

When does the two-sided limit $\lim_{x \to c} f(x)$ exist?

a. If and only if both one-sided limits exist and are equal ($\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L$)

b. Whenever $f(c)$ exists

c. Whenever $\lim_{x \to c^+} f(x)$ exists

d. If f is defined on $[c, \infty)$

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Graphical Limits and One-Sided Limits

For the step function $f(x)$ defined as 2 for $x < 1$ and 5 for $x \geq 1$, find $\lim_{x \to 1^-} f(x)$ and $\lim_{x \to 1^+} f(x)$.

a. Left limit $= 2$, Right limit $= 5$

b. Left limit $= 5$, Right limit $= 2$

c. Both equal 2

d. Both equal 5

26 [Foundational] • Target: Graphical Limits and One-Sided Limits

Does the two-sided limit $\lim_{x \to 1} f(x)$ exist for the step function in the previous question?

a. No, because the left limit (2) does not equal the right limit (5)

b. Yes, it equals 5

c. Yes, it equals 3.5

d. Yes, it equals 2

27 [Foundational] • Target: Algebraic Evaluation of Limits

Evaluate $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

a. 8

b. 0

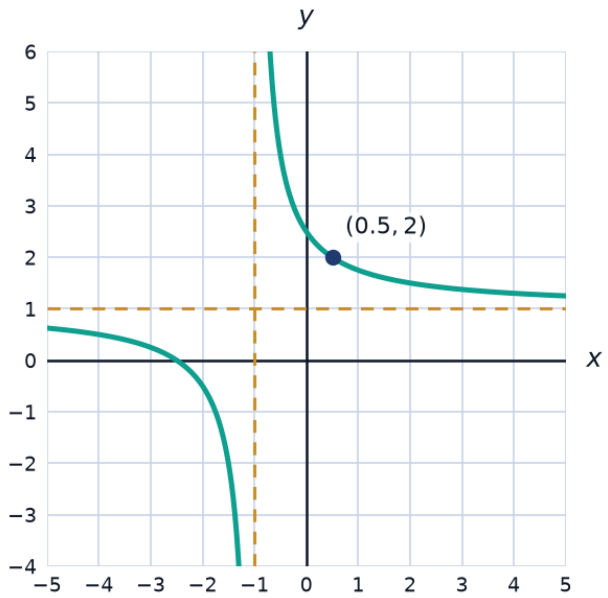
c. 4

d. Does not exist

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Algebraic Evaluation of Limits

Evaluate $\lim_{x \to 0} \frac{\sqrt{x+9}-3}{x}$.



a. 1/6

b. 1/3

c. 0

d. 1/9

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Algebraic Evaluation of Limits

Evaluate $\lim_{x \to -2} \frac{x^3+8}{x+2}$.

a. 12

b. 0

c. 4

d. 8

30 [Intermediate] • Target: Limits Involving Infinity

Evaluate $\lim_{x \to \infty} \frac{5x^2-2x}{2x^2+7x-1}$.

a. 5/2

b. ∞

c. 0

d. −2

© ClipSAT — clipsat.org | All Rights Reserved

31 **[Intermediate]** • **Target: Limits Involving Infinity**

Evaluate $\lim_{x\to\infty} \frac{3x+1}{x^2+5}$.

a. 0

b. 3

c. ∞

d. 1/5

32 **[Intermediate]** • **Target: Limits Involving Infinity**

Evaluate $\lim_{x\to 3^+} \frac{1}{x-3}$ and $\lim_{x\to 3^-} \frac{1}{x-3}$.

a. $+\infty$ and $-\infty$

b. $+\infty$ and $+\infty$

c. $-\infty$ and $+\infty$

d. 0 and 0

33 **[Intermediate]** • **Target: Limits Involving Infinity**

Using the limit definition $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$, find $f'(x)$ for $f(x) = 4x^2 - x + 4$.

a. $4x^2 - 1$

b. $8x + 4$

c. $4x - 1$

d. $8x - 1$

34 **[Intermediate]** • **Target: Evaluating limits**

Using the limit definition $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$, find $f'(x)$ for $f(x) = 3x^2 - 3x + 3$.

a. $6x - 3$

b. $6x + 3$

c. $3x^2 - 3$

d. $3x - 3$

35 **[Intermediate]** • **Target: One-sided limits and limits at infinity**

Using the limit definition $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$, find $f'(x)$ for $f(x) = 3x^2 + 4x + 5$.

a. $3x + 4$

b. $3x^2 + 4$

c. $6x + 5$

d. $6x + 4$

36 **[Intermediate]** • **Target: More limit computations**

Using the limit definition $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$, find $f'(x)$ for $f(x) = 3x^2 - 2x + 5$.

a. $3x - 2$

b. $6x - 2$

c. $3x^2 - 2$

d. $6x + 5$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Continuity and limits together

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 2x^2 + 2x - 1$.

a. $4x + 2$

b. $4x - 1$

c. $2x^2 + 2$

d. $2x + 2$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Limits from graphs and tables • Cross-Lesson: 5.01 Vectors: Components and Operations

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 4x^2 + 2x - 4$.

a. $4x^2 + 2$

b. $8x - 4$

c. $8x + 2$

d. $4x + 2$

39 [Challenging] • Target: Visualizing a limit graphically • Cross-Lesson: 5.02 The Dot Product

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 1x^2 + 4x - 5$.

a. $1x + 4$

b. $1x^2 + 4$

c. $2x + 4$

d. $2x - 5$

40 [Challenging] • Target: The formal idea of a limit (intuitive) • Cross-Lesson: 5.03 Parametric Equations

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 2x^2 - x + 0$.

a. $2x - 1$

b. $2x^2 - 1$

c. $4x + 0$

d. $4x - 1$

41 [Challenging] • Target: Limit laws • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 1x^2 - 3x - 1$.

a. $2x - 3$

b. $1x - 3$

c. $2x - 1$

d. $1x^2 - 3$

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Limits involving piecewise functions • Cross-Lesson: 6.02 Polar Equations and Graphs

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 4x^2 + x - 3$.

a. $4x^2 + 1$

b. $8x - 3$

c. $4x + 1$

d. $8x + 1$

43 [Challenging] • Target: Intuitive Concept and Numerical Estimates • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 3x^2 + x - 5$.

a. $6x - 5$

b. $6x + 1$

c. $3x + 1$

d. $3x^2 + 1$

44 [Challenging] • Target: Graphical Limits and One-Sided Limits • Cross-Lesson: 7.02 Logarithms and Their Properties

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 3x^2 - 4x + 3$.

a. $3x - 4$

b. $3x^2 - 4$

c. $6x + 3$

d. $6x - 4$

45 [Challenging] • Target: Algebraic Evaluation of Limits • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 1x^2 + 3x - 5$.

a. $1x^2 + 3$

b. $2x - 5$

c. $1x + 3$

d. $2x + 3$

© ClipSAT — clipsat.org | All Rights Reserved